Biochemistry Cumulative Examination

September 30, 2011

Answer the following questions about a recent departmental seminar, by Scott Feller from Wabash College.

1. Dr. Feller discussed the interaction of proteins with the lipids in cell membranes. What protein did he use in his work? 10 points

2. The lipids he spent the most time discussing were so-called ω-3 fatty acids. What specific structural feature(s) make(s) a fatty acid an ω-3? 10 points

3. Draw a typical saturated fatty acid and a typical polyunsaturated fatty acid. 20 points

4. Dr. Feller described the dynamic behavior of phospholipids containing a certain ω-3 fatty acid, docosahexaenoic acid or DHA. In particular, how was the DHA's behavior different from the other fatty acid in the phospholipid? Be sure to describe how each interacted with the protein. 30 points

5. How does the presence of DHA containing phospholipids affect the protein's properties and enzymatic activity? 10 points

6. How did Dr. Feller figure all this out? In other words, what techniques are used in his research group and by some of his collaborators? 15 points

7. What dietary sources are especially high in DHA? 5 points